Omar Contenedores

Command Prompt

```
C:\Users\USUARIO>docker version
Client:
 Version:           29.2.1
 API version:       1.53
 Go version:        go1.25.6
 Git commit:        a5c7197
 Built:             Mon Feb  2 17:20:16 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.62.0 (219486)
 Engine:
  Version:          29.2.1
  API version:      1.53 (minimum version 1.44)
  Go version:       go1.25.6
  Git commit:       6bc6209
  Built:            Mon Feb  2 17:17:24 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0

C:\Users\USUARIO>docker info
```

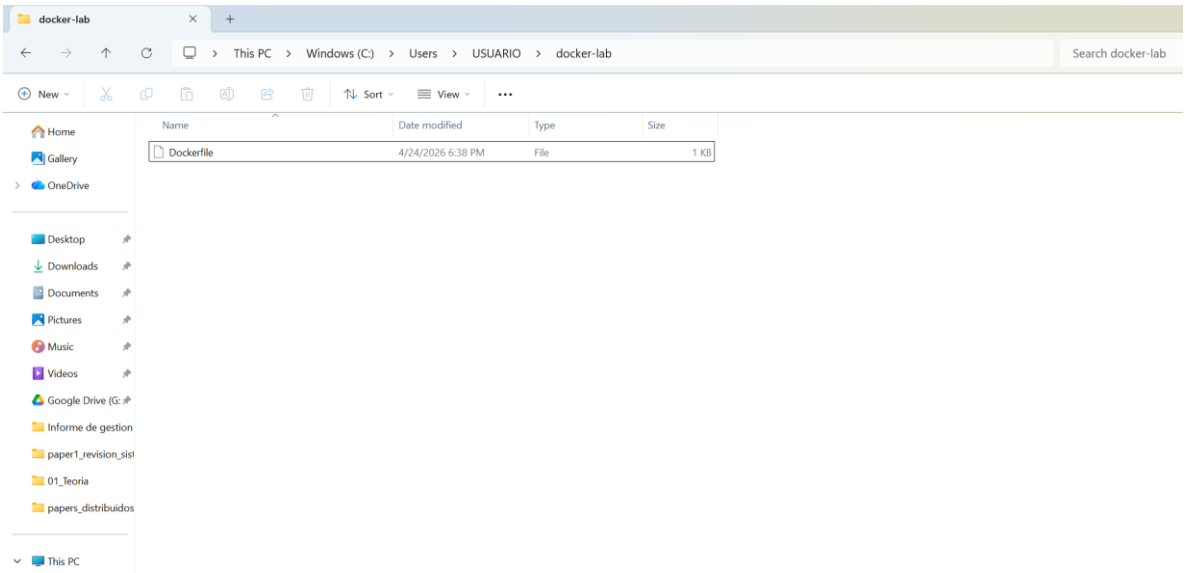
```
C:\Users\USUARIO>docker pull nginx:latest
latest: Pulling from library/nginx
ff048f1f2159: Pull complete
677c63196868: Pull complete
4677c2a9a3d4: Pull complete
3531af2bc2a9: Pull complete
ce776bbcda0d: Pull complete
801a1ad15b4e: Pull complete
85c66128325a: Pull complete
54c38c75806e: Download complete
69989ccd189b: Download complete
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

C:\Users\USUARIO>docker images
                                                                  i Info →   U  In Use
IMAGE                                    ID             DISK USAGE   CONTENT SIZE   EXTRA
nginx:latest                             6e23479198b9        240MB         65.8MB
penpotapp/backend:latest                 dbf308a07b19        1.3GB          409MB    U
penpotapp/exporter:latest                b0324003d10d       2.41GB          627MB    U
penpotapp/frontend:latest                caefaf8ef815        1.5GB          472MB    U
postgres:15                              0f989d656f95        633MB          164MB    U
postgres:15-alpine                       fceb6f86328c        392MB          110MB    U
sj26/mailcatcher:latest                  5d153a4daadf        211MB         66.1MB    U
us-central1-docker.pkg.dev/terence-unal/terence-images/api:latest
                                         a8e7d62e2afa        654MB          139MB
us-central1-docker.pkg.dev/terence-unal/terence-images/terence-api:latest
                                         0a6415a04130        654MB          139MB
us-central1-docker.pkg.dev/terence-unal/terence-images/web:latest
                                         deb27b7d4543        380MB         91.3MB
valkey/valkey:8.1                        49ccaa10c115        163MB           43MB    U

C:\Users\USUARIO>
```

```
C:\Users\USUARIO>mkdir docker-lab

C:\Users\USUARIO>cd docker-lab
```




```
C:\Users\USUARIO\docker-lab>docker build -t miapp:v1 .
[+] Building 5.3s (8/8) FINISHED                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                0.1s
 => => transferring dockerfile: 149B                                                0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                      2.0s
 => [auth] library/alpine:pull token for registry-1.docker.io                       0.0s
 => [internal] load .dockerignore                                                   0.0s
 => => transferring context: 2B                                                     0.0s
 => [1/3] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2  1.7s
 => => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2  0.0s
 => => sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd9f98b5 3.63MB / 3.63MB  1.4s
 => => extracting sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd9f98b50d2e1  0.2s
 => [2/3] WORKDIR /app                                                              0.1s
 => [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt                           0.6s
 => exporting to image                                                              0.5s
 => => exporting layers                                                             0.2s
 => => exporting manifest sha256:8934274936dc8704666bdd84a9fba36ed756770bf225e86  0.0s
 => => exporting config sha256:92025bd48f21844a2bc1c879f617f0cda66c2d50190a24140  0.0s
 => => exporting attestation manifest sha256:016c0c2e42e81798c28ef5948c6a07caf09  0.0s
 => => exporting manifest list sha256:8a928f73bfb09c479f9d8483fbd6b0a1e11775db27  0.0s
 => => naming to docker.io/library/miapp:v1                                         0.0s
 => => unpacking to docker.io/library/miapp:v1                                      0.1s

C:\Users\USUARIO\docker-lab>docker run --rm miapp:v1
Hola desde Docker

C:\Users\USUARIO\docker-lab>
```

```
C:\Users\USUARIO\docker-lab>docker run --rm miapp:v1
Hola desde Docker

C:\Users\USUARIO\docker-lab>docker tag miapp:v1 miapp:latest

C:\Users\USUARIO\docker-lab>docker images | grep miapp
'grep' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\USUARIO\docker-lab>docker images | findstr miapp
WARNING: This output is designed for human readability. For machine-readable output, pl
ease use --format.
miapp:latest                                                                    8a928f73bfb
0       12.1MB         3.63MB
miapp:v1                                                                        8a928f73bfb
0       12.1MB         3.63MB

C:\Users\USUARIO\docker-lab>
```

```
C:\Users\USUARIO\docker-lab>docker images | grep miapp
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miapp:latest                                                                    8a928f73bfb
0       12.1MB         3.63MB
miapp:v1                                                                        8a928f73bfb
0       12.1MB         3.63MB

C:\Users\USUARIO\docker-lab>docker image inspect miapp:v1 --format "{{.Id}}"
sha256:8a928f73bfb09c479f9d8483fbd6b0a1e11775db27953c508f7b2ad9a5f20df9

C:\Users\USUARIO\docker-lab>docker save -o miapp.tar miapp:v1

C:\Users\USUARIO\docker-lab>dir miapp.tar
 Volume in drive C is Windows
 Volume Serial Number is 98E2-DF34

 Directory of C:\Users\USUARIO\docker-lab

04/24/2026  06:43 PM         3,647,488 miapp.tar
               1 File(s)      3,647,488 bytes
               0 Dir(s)  586,077,827,072 bytes free

C:\Users\USUARIO\docker-lab>docker rmi miapp:v1
Untagged: miapp:v1

C:\Users\USUARIO\docker-lab>docker load -i miapp.tar
Loaded image: miapp:v1

C:\Users\USUARIO\docker-lab>docker images | findstr miapp
WARNING: This output is designed for human readability. For machine-readable output, pl
ease use --format.
miapp:latest                                                                    8a928f73bfb
0       12.1MB         3.63MB
miapp:v1                                                                        8a928f73bfb
0       12.1MB         3.63MB

C:\Users\USUARIO\docker-lab>
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miapp:latest                                                                      8a928f73bfb
0       12.1MB         3.63MB
miapp:v1                                                                          8a928f73bfb
0       12.1MB         3.63MB

C:\Users\USUARIO\docker-lab>docker rmi miapp:latest
Untagged: miapp:latest

C:\Users\USUARIO\docker-lab>docker images | findstr miapp
WARNING: This output is designed for human readability. For machine-readable output, pl
ease use --format.
miapp:v1                                                                          8a928f73bfb
0       12.1MB         3.63MB

C:\Users\USUARIO\docker-lab>docker image prune -f
Total reclaimed space: 0B

C:\Users\USUARIO\docker-lab>
```

Seccion 2-Contenedores

```
C:\Users\USUARIO\docker-lab>docker image prune -f
Total reclaimed space: 0B

C:\Users\USUARIO\docker-lab>docker pull nginx:latest
latest: Pulling from library/nginx
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest

C:\Users\USUARIO\docker-lab>docker run -d --name web1 nginx:latest
e86fff83f42b99c5c22cd050c11b3f7b4c5c04420f1f435b22de1a2b6b8e9cfe

C:\Users\USUARIO\docker-lab>docker ps
CONTAINER ID   IMAGE                       COMMAND                  CREATED         STA
TUS                 PORTS                                           NAMES
e86fff83f42b   nginx:latest                "/docker-entrypoint.…"   4 seconds ago   Up
4 seconds           80/tcp                                          web1
fc9bc4216841   postgres:15-alpine          "docker-entrypoint.s…"   3 weeks ago     Up
9 hours             0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp   terence-postgres
f66c164f1891   penpotapp/frontend:latest   "/bin/bash /entrypoi…"   5 weeks ago     Up
9 hours             0.0.0.0:9001->8080/tcp, [::]:9001->8080/tcp   penpot-penpot-fronten
d-1
8e71e9bfa66f   penpotapp/backend:latest    "/bin/bash run.sh"       5 weeks ago     Up
9 hours                                                           penpot-penpot-backend
-1
06dd516554f6   penpotapp/exporter:latest   "node app.js"            5 weeks ago     Up
9 hours                                                           penpot-penpot-exporte
r-1
1bf9edc2d9af   postgres:15                 "docker-entrypoint.s…"   5 weeks ago     Up
9 hours (healthy)   5432/tcp                                       penpot-penpot-postgre
s-1
5e8b6c08fcf3   sj26/mailcatcher:latest     "mailcatcher --foreg…"   5 weeks ago     Up
9 hours             0.0.0.0:1080->1080/tcp, [::]:1080->1080/tcp   penpot-penpot-mailcat
ch-1
802e9f8ba0a2   valkey/valkey:8.1           "docker-entrypoint.s…"   5 weeks ago     Up
9 hours (healthy)   6379/tcp                                       penpot-penpot-valkey-
1
a07990448d5d   0236ee02dcbc                "/docker-entrypoint.…"   7 weeks ago     Up
9 hours             80/tcp                                          laughing_blackwell

C:\Users\USUARIO\docker-lab>docker run -d --name web2 -p 8080:80 nginx:latest
624ade3073d09cce667b0639cefc0bf927ffc64cf8f0402e66424b7a28b768f0

C:\Users\USUARIO\docker-lab>
```

```
C:\Users\USUARIO\docker-lab>docker run -d --name web2 -p 8080:80 nginx:latest
624ade3073d09cce667b0639cefc0bf927ffc64cf8f0402e66424b7a28b768f0

C:\Users\USUARIO\docker-lab>curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>

C:\Users\USUARIO\docker-lab>
```

```
<p><em>Thank you for using nginx.</em></p>
</body>
</html>

C:\Users\USUARIO\docker-lab>docker ps -a
CONTAINER ID   IMAGE                       COMMAND                  CREATED         STA
TUS                     PORTS                                           NAMES
624ade3073d0   nginx:latest                "/docker-entrypoint.…"   2 minutes ago   Up
2 minutes               0.0.0.0:8080->80/tcp, [::]:8080->80/tcp         web2
e86fff83f42b   nginx:latest                "/docker-entrypoint.…"   2 minutes ago   Up
2 minutes               80/tcp                                          web1
fc9bc4216841   postgres:15-alpine          "docker-entrypoint.s…"   3 weeks ago     Up
9 hours                 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp     terence-postgres
f66c164f1891   penpotapp/frontend:latest   "/bin/bash /entrypoi…"   5 weeks ago     Up
9 hours                 0.0.0.0:9001->8080/tcp, [::]:9001->8080/tcp     penpot-penpot-fro
ntend-1
8e71e9bfa66f   penpotapp/backend:latest    "/bin/bash run.sh"       5 weeks ago     Up
9 hours                                                                 penpot-penpot-bac
kend-1
06dd516554f6   penpotapp/exporter:latest   "node app.js"            5 weeks ago     Up
9 hours                                                                 penpot-penpot-exp
orter-1
1bf9edc2d9af   postgres:15                 "docker-entrypoint.s…"   5 weeks ago     Up
9 hours (healthy)       5432/tcp                                        penpot-penpot-pos
tgres-1
5e8b6c08fcf3   sj26/mailcatcher:latest     "mailcatcher --foreg…"   5 weeks ago     Up
9 hours                 0.0.0.0:1080->1080/tcp, [::]:1080->1080/tcp     penpot-penpot-mai
lcatch-1
802e9f8ba0a2   valkey/valkey:8.1           "docker-entrypoint.s…"   5 weeks ago     Up
9 hours (healthy)       6379/tcp                                        penpot-penpot-val
key-1
e14094a28c39   0236ee02dcbc                "/docker-entrypoint.…"   7 weeks ago     Exi
ted (255) 5 weeks ago   0.0.0.0:8080->80/tcp                            interesting_gauss
a07990448d5d   0236ee02dcbc                "/docker-entrypoint.…"   7 weeks ago     Up
9 hours                 80/tcp                                          laughing_blackwel
l
193d52d52a22   0236ee02dcbc                "/docker-entrypoint.…"   7 weeks ago     Exi
ted (0) 7 weeks ago                                                     optimistic_nash
2a01d222a2e6   0236ee02dcbc                "/docker-entrypoint.…"   7 weeks ago     Exi
ted (0) 7 weeks ago                                                     bold_leakey

C:\Users\USUARIO\docker-lab>
```

```
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/defaul
t.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/defa
ult.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/24 23:46:37 [notice] 1#1: using the "epoll" event method
2026/04/24 23:46:37 [notice] 1#1: nginx/1.29.8
2026/04/24 23:46:37 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/24 23:46:37 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/04/24 23:46:37 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/24 23:46:37 [notice] 1#1: start worker processes
2026/04/24 23:46:37 [notice] 1#1: start worker process 30
2026/04/24 23:46:37 [notice] 1#1: start worker process 31
2026/04/24 23:46:37 [notice] 1#1: start worker process 32
2026/04/24 23:46:37 [notice] 1#1: start worker process 33
2026/04/24 23:46:37 [notice] 1#1: start worker process 34
2026/04/24 23:46:37 [notice] 1#1: start worker process 35
2026/04/24 23:46:37 [notice] 1#1: start worker process 36
2026/04/24 23:46:37 [notice] 1#1: start worker process 37
2026/04/24 23:46:37 [notice] 1#1: start worker process 38
2026/04/24 23:46:37 [notice] 1#1: start worker process 39
2026/04/24 23:46:37 [notice] 1#1: start worker process 40
2026/04/24 23:46:37 [notice] 1#1: start worker process 41
2026/04/24 23:46:37 [notice] 1#1: start worker process 42
2026/04/24 23:46:37 [notice] 1#1: start worker process 43
2026/04/24 23:46:37 [notice] 1#1: start worker process 44
2026/04/24 23:46:37 [notice] 1#1: start worker process 45
2026/04/24 23:46:37 [notice] 1#1: start worker process 46
2026/04/24 23:46:37 [notice] 1#1: start worker process 47
2026/04/24 23:46:37 [notice] 1#1: start worker process 48
2026/04/24 23:46:37 [notice] 1#1: start worker process 49
2026/04/24 23:46:37 [notice] 1#1: start worker process 50
2026/04/24 23:46:37 [notice] 1#1: start worker process 51
2026/04/24 23:46:37 [notice] 1#1: start worker process 52
2026/04/24 23:46:37 [notice] 1#1: start worker process 53
172.17.0.1 - - [24/Apr/2026:23:47:03 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0"
"-"

C:\Users\USUARIO\docker-lab>
```

```
C:\Users\USUARIO\docker-lab>docker inspect web2 --format "{{.State.Status}}"
running

C:\Users\USUARIO\docker-lab>docker stop web2
web2

C:\Users\USUARIO\docker-lab>docker start web2
web2

C:\Users\USUARIO\docker-lab>docker ps
CONTAINER ID   IMAGE                       COMMAND                  CREATED         STA
TUS                 PORTS                                           NAMES
624ade3073d0   nginx:latest                "/docker-entrypoint.…"   3 minutes ago   Up
6 seconds           0.0.0.0:8080->80/tcp, [::]:8080->80/tcp         web2
e86fff83f42b   nginx:latest                "/docker-entrypoint.…"   3 minutes ago   Up
3 minutes           80/tcp                                          web1
fc9bc4216841   postgres:15-alpine          "docker-entrypoint.s…"   3 weeks ago     Up
9 hours             0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp     terence-postgres
f66c164f1891   penpotapp/frontend:latest   "/bin/bash /entrypoi…"   5 weeks ago     Up
9 hours             0.0.0.0:9001->8080/tcp, [::]:9001->8080/tcp     penpot-penpot-fronten
d-1
8e71e9bfa66f   penpotapp/backend:latest    "/bin/bash run.sh"       5 weeks ago     Up
9 hours                                                             penpot-penpot-backend
-1
06dd516554f6   penpotapp/exporter:latest   "node app.js"            5 weeks ago     Up
9 hours                                                             penpot-penpot-exporte
r-1
1bf9edc2d9af   postgres:15                 "docker-entrypoint.s…"   5 weeks ago     Up
9 hours (healthy)   5432/tcp                                        penpot-penpot-postgre
s-1
5e8b6c08fcf3   sj26/mailcatcher:latest     "mailcatcher --foreg…"   5 weeks ago     Up
9 hours             0.0.0.0:1080->1080/tcp, [::]:1080->1080/tcp     penpot-penpot-mailcat
ch-1
802e9f8ba0a2   valkey/valkey:8.1           "docker-entrypoint.s…"   5 weeks ago     Up
9 hours (healthy)   6379/tcp                                        penpot-penpot-valkey-
1
a07990448d5d   0236ee02dcbc                "/docker-entrypoint.…"   7 weeks ago     Up
9 hours             80/tcp                                          laughing_blackwell

C:\Users\USUARIO\docker-lab>
```

```
TUS                  PORTS                                          NAMES
624ade3073d0   nginx:latest                "/docker-entrypoint.…"   3 minutes ago   Up
6 seconds            0.0.0.0:8080->80/tcp, [::]:8080->80/tcp        web2
e86fff83f42b   nginx:latest                "/docker-entrypoint.…"   3 minutes ago   Up
3 minutes            80/tcp                                         web1
fc9bc4216841   postgres:15-alpine          "docker-entrypoint.s…"   3 weeks ago     Up
9 hours              0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp    terence-postgres
f66c164f1891   penpotapp/frontend:latest   "/bin/bash /entrypoi…"   5 weeks ago     Up
9 hours              0.0.0.0:9001->8080/tcp, [::]:9001->8080/tcp    penpot-penpot-fronten
d-1
8e71e9bfa66f   penpotapp/backend:latest    "/bin/bash run.sh"       5 weeks ago     Up
9 hours                                                             penpot-penpot-backend
-1
06dd516554f6   penpotapp/exporter:latest   "node app.js"            5 weeks ago     Up
9 hours                                                             penpot-penpot-exporte
r-1
1bf9edc2d9af   postgres:15                 "docker-entrypoint.s…"   5 weeks ago     Up
9 hours (healthy)    5432/tcp                                       penpot-penpot-postgre
s-1
5e8b6c08fcf3   sj26/mailcatcher:latest     "mailcatcher --foreg…"   5 weeks ago     Up
9 hours              0.0.0.0:1080->1080/tcp, [::]:1080->1080/tcp    penpot-penpot-mailcat
ch-1
802e9f8ba0a2   valkey/valkey:8.1           "docker-entrypoint.s…"   5 weeks ago     Up
9 hours (healthy)    6379/tcp                                       penpot-penpot-valkey-
1
a07990448d5d   0236ee02dcbc                "/docker-entrypoint.…"   7 weeks ago     Up
9 hours              80/tcp                                         laughing_blackwell

C:\Users\USUARIO\docker-lab>docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
Digest: sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.10
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ #
```

```
VERSION_ID=3.20.10
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit

C:\Users\USUARIO\docker-lab>docker pause web1
web1

C:\Users\USUARIO\docker-lab>docker inspect web1 --format "{{.State.Paused}}"
true

C:\Users\USUARIO\docker-lab>docker unpause web1
web1

C:\Users\USUARIO\docker-lab>docker inspect web1 --format "{{.State.Paused}}"
false

C:\Users\USUARIO\docker-lab>docker stop alpine1
alpine1

C:\Users\USUARIO\docker-lab>docker rm alpine1
alpine1

C:\Users\USUARIO\docker-lab>docker rm -f web1
web1

C:\Users\USUARIO\docker-lab>docker ps -a | findstr alpine1

C:\Users\USUARIO\docker-lab>
```

Volumenes

```
C:\Users\USUARIO\docker-lab>docker ps -a | findstr alpine1

C:\Users\USUARIO\docker-lab>docker volume create datos_nginx
datos_nginx

C:\Users\USUARIO\docker-lab>docker volume ls
DRIVER    VOLUME NAME
local     a29a4c3829bfab23468068f2cacb6ef0019854ea4333262dff34bb180b2630e8
local     datos_nginx
local     penpot_penpot_assets
local     penpot_penpot_postgres_v15

C:\Users\USUARIO\docker-lab>docker volume inspect datos_nginx
[
    {
        "CreatedAt": "2026-04-25T00:17:25Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
        "Name": "datos_nginx",
        "Options": null,
        "Scope": "local"
    }
]

C:\Users\USUARIO\docker-lab>docker run -d --name webvol -p 8081:80 -v datos_nginx:/usr/
share/nginx/html nginx:latest
427667534cc8cbbce9474f6946e418ac139eb96e216ae7068254c2c559ea3db0

C:\Users\USUARIO\docker-lab>echo ^<h1^>Hola volumen^</h1^> > index.html

C:\Users\USUARIO\docker-lab>docker cp index.html webvol:/usr/share/nginx/html/index.htm
l
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html

C:\Users\USUARIO\docker-lab>curl http://localhost:8081
<h1>Hola volumen</h1>

C:\Users\USUARIO\docker-lab>docker rm -f webvol
webvol

C:\Users\USUARIO\docker-lab>docker volume rm datos_nginx
datos_nginx
```

```
C:\Users\USUARIO\docker-lab>docker volume inspect datos_nginx
[
    {
        "CreatedAt": "2026-04-25T00:17:25Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
        "Name": "datos_nginx",
        "Options": null,
        "Scope": "local"
    }
]

C:\Users\USUARIO\docker-lab>docker run -d --name webvol -p 8081:80 -v datos_nginx:/usr/
share/nginx/html nginx:latest
427667534cc8cbbce9474f6946e418ac139eb96e216ae7068254c2c559ea3db0

C:\Users\USUARIO\docker-lab>echo ^<h1^>Hola volumen^</h1^> > index.html

C:\Users\USUARIO\docker-lab>docker cp index.html webvol:/usr/share/nginx/html/index.htm
l
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html

C:\Users\USUARIO\docker-lab>curl http://localhost:8081
<h1>Hola volumen</h1>

C:\Users\USUARIO\docker-lab>docker rm -f webvol
webvol

C:\Users\USUARIO\docker-lab>docker volume rm datos_nginx
datos_nginx

C:\Users\USUARIO\docker-lab>docker volume ls
DRIVER    VOLUME NAME
local     a29a4c3829bfab23468068f2cacb6ef0019854ea4333262dff34bb180b2630e8
local     penpot_penpot_assets
local     penpot_penpot_postgres_v15

C:\Users\USUARIO\docker-lab>
```

Redes

```
C:\Users\USUARIO\docker-lab>docker network ls
NETWORK ID     NAME            DRIVER    SCOPE
20d25d9fc657   bridge          bridge    local
bdadadd4084a   host            host      local
5654f9f5d635   none            null      local
4583dc65b847   penpot_penpot   bridge    local

C:\Users\USUARIO\docker-lab>docker network create mi_red
f3779609fc0e6c8b4fc04b5f1f6d785e22a6e83f8603cea7071ae5cf3f12c783

C:\Users\USUARIO\docker-lab>docker network ls
NETWORK ID     NAME            DRIVER    SCOPE
20d25d9fc657   bridge          bridge    local
bdadadd4084a   host            host      local
f3779609fc0e   mi_red          bridge    local
5654f9f5d635   none            null      local
4583dc65b847   penpot_penpot   bridge    local

C:\Users\USUARIO\docker-lab>docker network inspect mi_red
[
    {
        "Name": "mi_red",
        "Id": "f3779609fc0e6c8b4fc04b5f1f6d785e22a6e83f8603cea7071ae5cf3f12c783",
        "Created": "2026-04-25T00:19:56.130081311Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
```

```
        "Name": "mi_red",
        "Id": "f3779609fc0e6c8b4fc04b5f1f6d785e22a6e83f8603cea7071ae5cf3f12c783",
        "Created": "2026-04-25T00:19:56.130081311Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.19.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }


:\Users\USUARIO\docker-lab>
```

```
C:\Users\USUARIO\docker-lab>docker network ls
NETWORK ID     NAME            DRIVER    SCOPE
20d25d9fc657   bridge          bridge    local
bdadadd4084a   host            host      local
f3779609fc0e   mi_red          bridge    local
5654f9f5d635   none            null      local
4583dc65b847   penpot_penpot   bridge    local

C:\Users\USUARIO\docker-lab>docker network inspect mi_red
[
    {
        "Name": "mi_red",
        "Id": "f3779609fc0e6c8b4fc04b5f1f6d785e22a6e83f8603cea7071ae5cf3f12c783",
        "Created": "2026-04-25T00:19:56.130081311Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
```

ENG 7:20

```
C:\Users\USUARIO\docker-lab>docker run -d --name caja1 --network mi_red alpine:3.20 sle
ep 300
526b40e347211f1d0a999024d558f16d89e722d647272f4e2f7abed1ff8b49bb

C:\Users\USUARIO\docker-lab>docker run -d --name caja2 --network mi_red alpine:3.20 sle
ep 300
edfe0072c0de135c9476228f4c56fb76567c5385d307ad6af903dc2afea3067d

C:\Users\USUARIO\docker-lab>docker exec caja1 ping -c 2 caja2
PING caja2 (172.19.0.3): 56 data bytes
64 bytes from 172.19.0.3: seq=0 ttl=64 time=0.802 ms
64 bytes from 172.19.0.3: seq=1 ttl=64 time=0.087 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.087/0.444/0.802 ms

C:\Users\USUARIO\docker-lab>docker network disconnect mi_red caja2

C:\Users\USUARIO\docker-lab>docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'

C:\Users\USUARIO\docker-lab>
```

Docker compose

```
C:\Users\USUARIO\docker-lab>notepad compose.yml

C:\Users\USUARIO\docker-lab>docker compose up -d
[+] up 2/2
 ✔Network docker-lab_default Created                                        0.0s
 ✔Container docker-lab-web-1 Created                                        0.1s

C:\Users\USUARIO\docker-lab>docker compose ps
NAME               IMAGE          COMMAND                  SERVICE   CREATED         ST
ATUS         PORTS
docker-lab-web-1   nginx:latest   "/docker-entrypoint.…"   web       7 seconds ago   Up
 7 seconds   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp

C:\Users\USUARIO\docker-lab>curl http://localhost:8082
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>

C:\Users\USUARIO\docker-lab>
```

```
C:\Users\USUARIO\docker-lab>docker compose logs
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to per
form configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-def
ault.sh
web-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf
.d/default.conf
web-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/co
nf.d/default.conf
web-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates
.sh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes
.sh
web-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1  | 2026/04/25 00:23:30 [notice] 1#1: using the "epoll" event method
web-1  | 2026/04/25 00:23:30 [notice] 1#1: nginx/1.29.8
web-1  | 2026/04/25 00:23:30 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1  | 2026/04/25 00:23:30 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
web-1  | 2026/04/25 00:23:30 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker processes
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 29
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 30
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 31
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 32
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 33
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 34
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 35
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 36
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 37
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 38
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 39
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 40
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 41
```

```
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 46
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 47
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 48
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 49
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 50
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 51
web-1  | 2026/04/25 00:23:30 [notice] 1#1: start worker process 52
web-1  | 172.20.0.1 - - [25/Apr/2026:00:23:42 +0000] "GET / HTTP/1.1" 200 896 "-" "curl
/8.18.0" "-"

C:\Users\USUARIO\docker-lab>docker compose down
[+] down 2/2
 ✔Container docker-lab-web-1 Removed                                          0.3s
 ✔Network docker-lab_default Removed                                          0.2s

C:\Users\USUARIO\docker-lab>docker compose ps
NAME      IMAGE     COMMAND   SERVICE   CREATED   STATUS    PORTS

C:\Users\USUARIO\docker-lab>
```

Escalar servicio

```
C:\Users\USUARIO\docker-lab>docker compose down
[+] down 2/2
 ✔Container docker-lab-web-1 Removed                                        0.3s
 ✔Network docker-lab_default Removed                                        0.2s

C:\Users\USUARIO\docker-lab>docker compose ps
NAME      IMAGE     COMMAND   SERVICE   CREATED   STATUS    PORTS

C:\Users\USUARIO\docker-lab>notepad compose.yml

C:\Users\USUARIO\docker-lab>docker compose up -d --scale web=3
[+] up 4/4
 ✔Network docker-lab_default Created                                        0.0s
 ✔Container docker-lab-web-3 Created                                        0.1s
 ✔Container docker-lab-web-2 Created                                        0.1s
 ✔Container docker-lab-web-1 Created                                        0.1s

C:\Users\USUARIO\docker-lab>docker compose ps
NAME                IMAGE          COMMAND                  SERVICE   CREATED         ST
ATUS         PORTS
docker-lab-web-1   nginx:latest   "/docker-entrypoint.…"   web       4 seconds ago   Up
 4 seconds   80/tcp
docker-lab-web-2   nginx:latest   "/docker-entrypoint.…"   web       4 seconds ago   Up
 3 seconds   80/tcp
docker-lab-web-3   nginx:latest   "/docker-entrypoint.…"   web       4 seconds ago   Up
 4 seconds   80/tcp

C:\Users\USUARIO\docker-lab>docker compose down
[+] down 4/4
 ✔Container docker-lab-web-2 Removed                                        0.5s
 ✔Container docker-lab-web-3 Removed                                        0.4s
 ✔Container docker-lab-web-1 Removed                                        0.5s
 ✔Network docker-lab_default Removed                                        0.3s

C:\Users\USUARIO\docker-lab>
```

Dockerfile Avanzado

```
C:\Users\USUARIO\docker-lab\dockerfile-test>docker build -t dockerfile-demo --build-arg
 VERSION=2.0 .
[+] Building 0.8s (10/10) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                   0.0s
 => => transferring dockerfile: 352B                                                   0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                         0.0s
 => [internal] load .dockerignore                                                      0.0s
 => => transferring context: 2B                                                        0.0s
 => [1/5] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2  0.0s
 => => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2  0.0s
 => [internal] load build context                                                      0.0s
 => => transferring context: 56B                                                       0.0s
 => CACHED [2/5] WORKDIR /app                                                          0.0s
 => [3/5] COPY archivo.txt /app/archivo.txt                                            0.0s
 => [4/5] RUN echo "version=2.0" > /app/version.txt                                    0.2s
 => [5/5] ADD archivo.txt /app/archivo2.txt                                            0.0s
 => exporting to image                                                                 0.3s
 => => exporting layers                                                                0.1s
 => => exporting manifest sha256:ca2da826cd266ee9a79522d7e780b41a59b05f241139878  0.0s
 => => exporting config sha256:cd27c915550beb33a07acc60827c88678d05008069a438aa9  0.0s
 => => exporting attestation manifest sha256:9698b8bd8da1f51a764defe3db1b5d02392  0.0s
 => => exporting manifest list sha256:75257071c860e3fd98bfd2a5ce6b8d6652c4fdf524  0.0s
 => => naming to docker.io/library/dockerfile-demo:latest                              0.0s
 => => unpacking to docker.io/library/dockerfile-demo:latest                           0.1s

C:\Users\USUARIO\docker-lab\dockerfile-test>docker run --rm dockerfile-demo
APP_ENV=dev
contenido local
version=2.0

C:\Users\USUARIO\docker-lab\dockerfile-test>docker image inspect dockerfile-demo --form
at "{{json .Config.Labels}}"
{"autor":"prueba"}

C:\Users\USUARIO\docker-lab\dockerfile-test>docker image inspect dockerfile-demo --form
at "{{json .Config.Env}}"
["PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin","APP_ENV=dev"]

C:\Users\USUARIO\docker-lab\dockerfile-test>docker image inspect dockerfile-demo --form
at "{{json .Config.Volumes}}"
{"/data":{}}

C:\Users\USUARIO\docker-lab\dockerfile-test>
```